Project 4 | OnShape | About 45 min

Spinning Top in OnShape

Draw **half** of a top, spin it around a line, and OnShape makes the whole 3D top for you!

The big idea: Revolve

A top is perfectly round. So instead of building it piece by piece, you draw the outline of **one half** (like a top cut in half), and the **Revolve** tool spins that drawing all the way around a center line. It's like a pottery wheel!

WHAT MAKES A TOP SPIN A LONG TIME?

- **Wide and heavy near the bottom**: a wide disc keeps spinning, like a flywheel.
- **A sharp, centered tip**: less rubbing on the table.
- **A short handle**: just long enough to twist with your fingers.

THE 3D PRINTER RULE

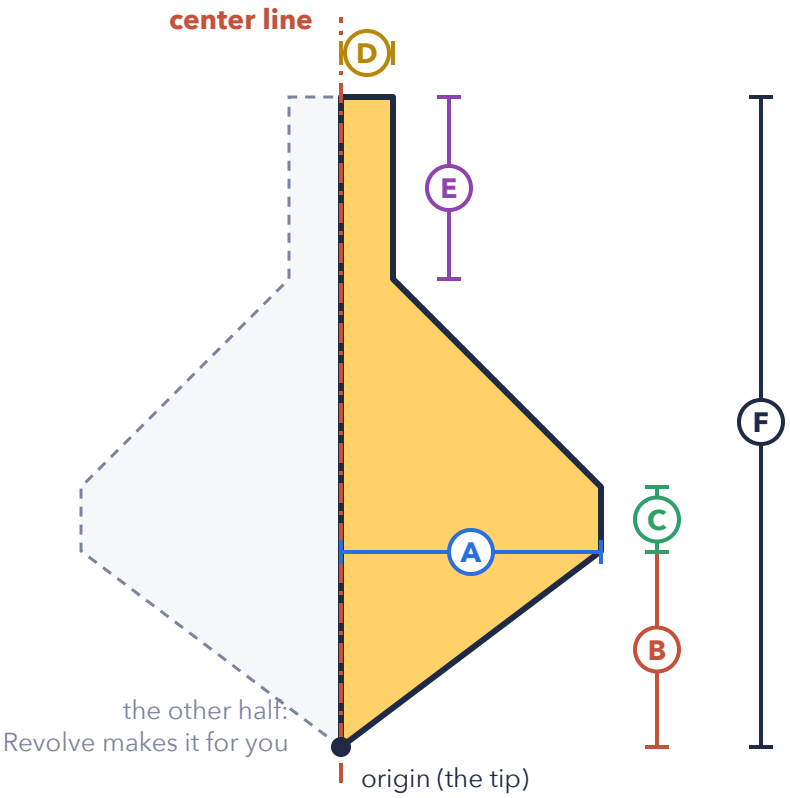
Your top prints **upside down**, handle on the bed. The printer can't print in mid-air, so slanted parts need to be **steep** (no flatter than 45°). That's why the starter design slants up to the handle instead of being flat.

BEFORE YOU START

You need your OnShape account (see **Getting Started with OnShape**). Make a new document named **Top – First Name Last Initial**, and set **Workspace units** to **Millimeter**.

The starter design

This is the half you'll draw. The red center line is where it spins. Each letter is a size you'll type in.



A	Disc, center to edge	20
B	Tip cone height	15
C	Disc edge	5

D	Handle, center to edge	4
E	Handle length	14
F	Total height	50

All sizes in mm. The finished top is 40 mm wide (2 × A) and 50 mm tall.

WHY ONLY HALF THE WIDTH?

You only draw from the center line out to the edge. When it spins around, the other half appears by itself. So A = 20 makes a disc that's **40 mm** across!

1 Start a sketch

- [ ] Click **Sketch** in the toolbar, then click the **Front** plane (in the graphics area or the feature list).
- [ ] OnShape turns the view so you're looking straight at the plane. If it doesn't, press N (it means "look straight at my sketch").

2 Draw the outline

Use the **Line** tool (L). Click once at each corner. Don't worry about exact sizes yet, just get the shape right!

- [ ] Click on the **origin** (the dot in the middle). That's the tip of your top.
- [ ] Click up and to the right (the slanted cone), then straight up a little (the disc edge).
- [ ] Click up and to the left (the slant toward the handle), then straight up (the handle).
- [ ] Click straight left, so you land right above the origin. Then click back **on the origin** to close the shape.
- [ ] Press Esc. If the inside of your shape turns **gray/shaded**, it's closed. 🎉 If not, a corner isn't connected: drag the loose end onto the point it should touch.

STRAIGHT LINES
When a line is perfectly up-down or left-right, OnShape shows a little **|** or **–** symbol next to it. You want those on the edge, handle, top and center lines!

3 Add the sizes

- [ ] Click **Dimension** (D). Click a line, move your mouse to place the number, and click. Type the size and press Enter.
- [ ] To size A and D: click the **center line**, then the edge line, and place it above.
- [ ] To size B: click the **origin**, then the corner at the end of the cone. Move your mouse to the **side** so the number measures up-and-down.
- [ ] Do all six: A 20, B 15, C 5, D 4, E 14, F 50. Then click the green ✔ to finish the sketch.

4 Revolve it!

- [ ] Click **Revolve** in the toolbar.
- [ ] For **Faces and sketch regions**, click inside your shaded shape.
- [ ] For **Revolve axis**, click the **center line** (the line going straight up from the origin).
- [ ] Make sure it says **Full**, then click the green ✔. Spin the view around… it's a top! 🌀

UH OH?

Revolve won't work or looks weird? You probably picked the wrong line as the axis. Right-click **Revolve 1** in the feature list, choose **Edit**, and pick the center line again.

5 Round the sharp edges (optional)

- [ ] Click **Fillet**. Click the sharp edges around the disc. Set the radius to `1` mm and click ✔.
- [ ] Don't round the tip too much. A sharp tip spins longer!

6 Make it yours 🎨

Now experiment! Right-click **Sketch 1** → **Edit**, change a size, and click ✔. The whole top updates by itself.

- Make the disc wider or narrower (keep the whole top under `60` mm wide).
- Try a curvy outline with the **Tangent arc** tool instead of straight lines.
- Remember the printer rule: no slants flatter than 45°, and no overhanging ledges.

My top's sizes: A _____ B _____ C _____ D _____ E _____ F _____

7 Hand it in 📤

- [ ] Check the document name says **Top – First Name Last Initial**. Click it to fix it if needed.
- [ ] Click **Share** (top right), type your teacher's email, choose **Can view**, turn on **Export**, and click **Share**.
- [ ] Tell your teacher your top is ready to print! 🖨️

🏆 Spin-off challenge

When the tops are printed, we'll spin them and time them. Fill this in on spin day!

Spin	Time (seconds)	Notes (wobbly? fell over? walked around?)
1		
2		
3		

🤔 Think about it

1. Whose top spun the longest in class? What was different about its shape?

2. What's one thing that's easier in OnShape than in TinkerCad? What's harder?

DONE EARLY?
Try the **Box with a Lid** challenge!